

# Rhythmiq

*"Your Vibes, Managed"*

## Lab 1 Outline

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# 1. Introduction

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## 1.1 Ecosystem Context

Rhythmiq operates within the hospitality industry, specifically among small cafes, coffee shops, and local restaurants, and the same need shows up in other public spaces such as gyms. In these venues the atmosphere is a core part of what the customer is buying, and background music is one of the strongest tools for shaping it. Throughout this document, the team uses the word “vibe” to mean the mood of a place and the way it makes people feel [Cambridge Dictionary, 2021].

The research on music and customer behavior is consistent. A BMI/NRG study of 1,000 U.S. consumers aged 21 and older found that 80% would stay longer if they heard good music playing and 60% would buy more food and drinks if they heard music they liked [BMI, 2023]. An MRC Data study similarly reported that 41% of customers will spend more time and money in a business if they like the music, and that customers are 39% more likely to return to a business when they enjoy it [MRC Data, 2021]. Music is therefore closely tied to customer satisfaction, how long customers stay, and how much they spend.

## 1.2 Problem Scope

Although atmosphere matters, small cafes and restaurants struggle to manage their background music. The options open to them are either expensive and tied to specific hardware or offer little customization. Without an affordable way for customers to take part, staff end up handling song requests manually, which is inefficient during busy times, and owners can find themselves fielding complaints from several customers at once about what is playing. Poor music choices also carry a lasting cost, because they lead customers not to return, and repetitive playlists wear on regulars who do not want to hear the same songs over and over.

At the same time, businesses miss out on customer engagement and potential revenue, because there is no affordable, controlled, and legal way for customers to interact with the music. The problem touches three groups. Customers want influence over the atmosphere through a seamless, low-friction interaction. Owners and staff want increased engagement without staff interruptions. Stakeholders such as artists, record labels, and music platforms (for example iTunes and Spotify) are affected by how music is used and licensed in commercial spaces.

## 1.3 Current Solutions and Limitations

Small venues currently have four main options for managing their music, and each one falls short of what a small business needs.

**TouchTunes.** TouchTunes is the best-known customer-facing jukebox platform, but it depends on proprietary hardware that must be installed on site, and customers need an app or the machine itself to make selections. The high cost and the size of the hardware make it a fit for large bars and restaurants rather than for a thirty-seat cafe [TouchTunes, n.d.]. Bar owners discussing the service online also report keeping only about 33–39% of the revenue their machines generate [r/BarOwners, n.d.].

**Rockbot.** Rockbot offers a business music platform that lets venues manage playlists and take requests, but it is a high-cost system that tends to scare off small businesses, it recommends dedicated hardware, and it requires an app or machine [Rockbot, n.d.-a]. Its mobile access is limited or slower than customers want to use, so it does not give a small venue a truly smooth interactive experience. Its subscriber terms also list no revenue share for the business, which means the venue earns nothing from the engagement [Rockbot, n.d.-b].

**Manual staff requests.** In the absence of a system, many customers simply ask the staff to change the song. This approach costs no money, but it pulls employees away from their primary work, especially during peak hours when the line is longest. It also produces inconsistent results, since whether a request is honored depends on who is working and how busy they are, and it gives the owner no way to keep the music aligned with the venue's intended atmosphere.

**Generic personal streaming.** Some owners turn to consumer streaming accounts such as Spotify Free or Spotify Personal because they are cheap and familiar. These services are licensed for personal listening only, so using them to play music in a business without a commercial license is not permitted and exposes the owner to legal risk. They also provide no mechanism for customers to interact with the music selection.

## 1.4 Market Gap

Taken together, these limitations reveal three gaps that Rhythmiq is designed to fill. The first is the "cost" gap: no affordable option works with the devices a small venue already owns, because TouchTunes and Rockbot both depend on proprietary hardware. The second is the "control" gap: owners may be concerned that customers will play a song that does not fit, so they need a way to let customers choose music without losing control of the atmosphere. The third is the "staff" gap: there is no way for customers to make song requests without interrupting employees, which creates what the team calls "social friction" between customers and staff.

This gap has consequences for both sides of the counter. Customers have no way to influence the atmosphere of a venue digitally, even though nearly everyone already carries a smartphone. Owners, meanwhile, are missing a potential secondary revenue stream, because the existing competitors return between none and roughly 39% of the revenue to the business and most small venues have no music engagement to monetize at all.

## 1.5 Required Solution Characteristics

A solution that closes these gaps must meet several requirements. First, it must be low cost and viable on a small business budget, which rules out large upfront purchases. Second, it must be hardware agnostic and work with equipment the venue already has. Third, it must let customers request songs without interrupting staff, so that the burden of song requests moves off employees entirely. Fourth, it must comply with commercial music licensing requirements administered by ASCAP, BMI, and SESAC, so that owners can use it without legal exposure. Fifth, it must give the owner and staff control over the final song selection, so that the venue keeps a curated vibe that matches its atmosphere. Finally, it

should share a high portion of any revenue it generates with the business, so that small venues have a real incentive to adopt it.

*[Insert Figure 1 here]*

**Figure 1:** Current process flow. A customer has a song request and asks staff to play it. Staff must decide whether they are capable of playing the specific song, since the system may not accommodate song additions and this uses employee time inefficiently, and then whether the song is acceptable to the business, which the employee or owner must determine. The song is then either played or not played.

Rhythmiq is designed to meet each of these requirements. It is a web-based, hardware-agnostic jukebox platform for local spaces, summarized by the tagline “Your Vibes, Managed.” The remainder of this document describes the product, the features and components that make it work, and a case study demonstrating how it would be used in a real venue.

## 2. Rhythmiq Product Description

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### 2.1 Overview

Rhythmiq is a software-only solution that replaces expensive proprietary hardware and manual staff processes with a seamless digital music ecosystem for small local spaces. Customers scan a QR code on their table to browse and request songs from their own phones, the venue's owner or staff decide what kinds of music are allowed, and licensed audio plays through the equipment the business already has. The product was originally called VibeQueue, but the team renamed it after learning that another app had launched under the same name in March 2026.

### 2.2 Goals and Objectives

Rhythmiq has five primary goals. The first is to give customers a frictionless way to influence the venue's music through their own smartphones, with no app download standing in the way. The second is to give owners and staff control over which songs can be requested, so that the atmosphere is always protected. The third is to provide an optional revenue stream through a pay-to-request model in which the business owner receives the majority of the revenue. The fourth is to ensure that all music playback is legally licensed for commercial use. The fifth is to make the platform deployable on hardware the business already owns, keeping the barrier to adoption as low as possible.

The team has also defined two limits on what the product will do. Rhythmiq will not generate custom songs for customers at this time, and it will not play any music that the business owner does not have the proper licensing to play.

### 2.3 Unique Differentiators vs. Competition

Rhythmiq differs from existing options in four main ways. First, it is software-only and hardware agnostic, running on devices the business already owns, whereas TouchTunes and Rockbot both require an app or a machine. Second, it pairs a low cost of entry with an industry-leading 67% revenue share for the business owner, compared with roughly 33–39% for TouchTunes and 0% for Rockbot. Third, it offers a customer-facing QR interface for song requests rather than requiring staff to mediate them, which removes the interruption problem described earlier. Fourth, it is legally compliant through SoundMachine, which holds ASCAP, BMI, and SESAC licensing, in contrast to unlicensed personal streaming.

### 2.4 System Flow

The system flow begins when a customer scans a QR code and is taken to the Web App UI in their mobile browser. When the customer searches for a song, the search is passed to the Vibe Engine, which retrieves the song's metadata from the Soundcharts API and checks it against the vibe constraints stored in the Venue Database. If the song passes validation, the Vibe Engine sends the validated song selection to the Local Server and sends the audio signal to SoundMachine for playback through the venue's speakers. SoundMachine reports its playback status back to the Vibe Engine, and the Vibe Engine logs

each song that plays in the Venue Database. Songs that fail validation are rejected before they ever reach the queue.

Two other flows run alongside this one. When a customer chooses to pay for priority, the Vibe Engine sends a payment request to the Stripe API. Separately, the venue owner sets the venue's vibe constraints and updates them over time, and those settings are stored in the Venue Database where the Vibe Engine reads them for every request. Figure 2 shows this flow as a Major Functional Components Diagram (MFCD), organized into presentation, application, and data layers.

*[Insert Figure 2 here]*

**Figure 2:** Rhythmiq major functional components diagram (MFCD). The customer and venue owner interact through the Web App UI (presentation layer). The Vibe Engine connects to the Soundcharts API, Stripe API, SoundMachine, and Local Server (application layer) and to the Venue Database (data layer).

## 2.5 Key Product Features and Capabilities

### 2.5.1 QR Code Access

Customers scan a QR code placed on their table to open the venue's Rhythmiq portal in a mobile browser. No app download is required, and no login is required beyond a simple session, which removes the biggest source of friction for a casual, one-time visitor and keeps the personal data collected to a minimum. Each business receives its QR code when the owner creates a venue account, and the code links the customer's device to that specific business's portal, so a single platform can serve many venues while keeping each venue's settings and queue separate.

### 2.5.2 Song Browsing and Requests

Once inside the portal, customers can search and browse for songs, which are displayed by genre and vibe. Searching and song details are powered by the Soundcharts API, which supplies each song's metadata. Results are filtered in real time based on the venue's active vibe constraints, so customers only see songs that the venue would actually allow, and results are designed to load quickly. This keeps the experience smooth, since customers are not tempted to request songs that would be rejected, and it reinforces the venue's identity through the choices on offer. When a customer selects a song, the request is validated against the business's filters, added to the queue, and confirmed on screen.

### 2.5.3 Vibe Engine (Custom Song Selection)

The Vibe Engine is a proprietary matching algorithm built by the Rhythmiq team, and it sits at the center of the system. It evaluates each requested song against the venue's allowed genre, tempo, mood, and popularity settings, making song recommendations that fit the venue's music preferences, and it rejects non-matching requests before they reach the queue. This component is significant for two reasons. Legally, it replaces the need for the Spotify API, which carries strict commercial use restrictions, with a safe in-house solution built on properly licensed metadata. From the venue's perspective, it allows the owner to maintain the brand atmosphere even while accepting input from customers.

#### **2.5.4 Business Control Center**

The Business Control Center is the dedicated dashboard for owners and staff and serves as the core management interface for the platform. Owners can filter songs by genre and mood, set genre allowlists and blocklists, and automatically block explicit content. Owners and staff can manage the live song queue, and staff can reorder songs, remove them, or block inappropriate requests so that disruptive requests never play. User management lets them deal with problem users as well. The dashboard also tracks revenue, showing live updates from song requests with a comparison to the previous week and year, along with statistical metrics that help owners understand what their customers want. Access is divided by role: owners configure filters, manage queue rules, and view revenue but cannot bypass licensing rules, while staff manage the queue and assist customers but cannot change business settings.

#### **2.5.5 Dynamic Vibe Moderation**

Dynamic Vibe Moderation applies integrated content filters across the entire system. Owners configure the genre and mood limits, tempo range, and explicit content rules once, and the system enforces those settings automatically on every request, rejecting disallowed songs without any staff involvement. Because enforcement is automatic, staff never need to review or police individual requests, and the venue's atmosphere stays consistent throughout the day even as different customers come and go. Owners can also adjust the rules over time, for example to shift from a calm morning profile to a livelier evening one.

#### **2.5.6 Pay to Request**

Pay to Request is an optional monetization feature. Customers can pay a small fee, for example between \$0.99 and \$1.99, to boost a song higher in the queue, with one or more priority tiers available, and the payment is processed automatically through Stripe. Payments are recorded for each request, and the business owner receives 67% of the revenue, which is well above what competing platforms offer. Rhythmiq relies on a trusted payment system and does not store card data locally, which limits payment and security risk for both the venue and the customer. This feature turns music engagement from a cost of doing business into a source of income for the owner.

#### **2.5.7 Licensed Streaming via SoundMachine**

All audio playback is routed through SoundMachine, a commercial streaming service that holds the necessary ASCAP, BMI, and SESAC licenses. Routing playback this way ensures that Rhythmiq is fully legal for business use and removes the compliance burden from the owner, who would otherwise need to arrange licensing separately. Rhythmiq never plays music directly from unlicensed sources.

#### **2.5.8 Hardware-Agnostic Deployment**

Rhythmiq runs on devices the business already owns, including staff phones, existing tablets, or a dedicated laptop. No proprietary console needs to be purchased, and the venue's existing speakers are used for playback. This keeps the cost of adoption low and lets a business begin using the platform as soon as it signs up.

## 2.6 Major Components (Hardware/Software)

### 2.6.1 Hardware Requirements

Rhythmiq requires no proprietary or dedicated hardware. The platform runs on any internet-connected device, such as a tablet, smartphone, or laptop or desktop computer, that the venue already owns. The only physical connection needed is between the venue's existing speaker system and the device that plays the audio, which most venues already have in place for their current music setup.

### 2.6.2 Software Architecture

The software architecture, illustrated in the MFCD in Figure 2, consists of seven major components organized into three layers. The Web App UI sits in the presentation layer, the Vibe Engine and the external services and local components it works with sit in the application layer, and the Venue Database sits in the data layer.

**Web App UI.** This is the interface that customers and venue owners use, reached through the QR code. It is a responsive web app where customers search for songs and request them, and where owners set and update their vibe settings through the Business Control Center.

**Vibe Engine.** This is the custom-built song selection and validation logic. It receives each song search from the Web App UI, gets song metadata from Soundcharts, reads the vibe constraints from the Venue Database, and returns a validated or rejected result. It passes validated selections to the Local Server, sends the audio signal to SoundMachine, receives playback status back, sends payment requests to Stripe, and logs each song that plays.

**Soundcharts API.** This third-party music analytics service supplies song metadata, including genre, tempo, mood, popularity, and an explicit flag, which is used for song lookup and vibe matching.

**Stripe API.** This third-party payment service processes the payment requests that customers make when they pay for priority, so that card data is handled by a trusted provider rather than stored by Rhythmiq.

**SoundMachine.** This third-party licensed commercial streaming service handles the actual audio playback in the venue, receives the audio signal from the Vibe Engine, reports playback status back, and ensures compliance with ASCAP, BMI, and SESAC.

**Local Server.** This component receives the validated song selections from the Vibe Engine at the venue, so that only approved songs move on toward playback.

**Venue Database.** This database stores each venue's vibe settings, such as genre rules, explicit flags, and tempo limits, along with the song queue state and transaction and revenue data. It supplies the constraints to song selection that the Vibe Engine checks on every request, and it receives the logged song playback and the venue owner's updated vibe settings.

In the prototype, every component is fully implemented except the Web App UI and the Stripe integration, which are only partially implemented. The Stripe integration cannot be completed because no bank account will be associated with the prototype. The team has not had to eliminate any major features and considers the remaining work achievable within its planned sprints.

### 2.6.3 Technologies, APIs, and Frameworks

Rhythmiq is written in Python, chosen for the team's familiarity and for the strength and popularity of the Django web framework. The project uses Django along with the Django REST Framework for its APIs, Django Channels for real-time updates such as live queue changes, the Requests library for calling external services, and the Stripe Python library for payments. Data is stored in a PostgreSQL database, and Docker is used to package the application. The system integrates the Soundcharts API for music metadata and the SoundMachine API for licensed audio playback, and it generates venue-specific QR codes that link customers to each business's portal. The web front end is designed mobile-first with a responsive layout, since nearly all customer interaction will happen on a phone.

For development, the team uses Visual Studio Code as its editor, GitHub for version control, and GitHub Actions for automated testing and deployment. One architectural note is worth recording. The initial prototype explored the Spotify API, but the team pivoted after finding that Spotify's commercial use licensing restrictions made it unsuitable for a business-facing product. The current architecture uses Soundcharts and SoundMachine instead, which eliminates that legal risk.

## 3. Identification of Case Study

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### 3.1 Who

The case study focuses on “Brewpoint Café,” a single-location independent coffee shop. The business is run by its owner, Jamie, along with two baristas, and has roughly thirty seats. Its customers are primarily morning commuters and remote workers, and its busiest hours are from 7 AM to 2 PM.

### 3.2 What Problem

During the morning rush, the baristas are interrupted ten to fifteen times by customers verbally requesting songs. Each interruption slows service at the moment when speed matters most, and requests from several customers at once can turn into complaints about what is playing. When staff try to be accommodating, the owner also loses control of the café’s intended acoustic and lo-fi vibe, since the music drifts toward whatever was requested last. On top of these problems, the café earns no revenue from music engagement, even though customers clearly care about what is playing.

### 3.3 Which Features Tested

The case study exercises the five features that matter most to a venue like Brewpoint. QR Code Access is tested through cards placed on each table. Song Browsing and Requests is tested through Soundcharts-powered search. The Business Control Center is configured with genre restrictions limited to acoustic, indie, and lo-fi, with explicit content blocked and a maximum tempo of 120 BPM. Dynamic Vibe Moderation is tested by having the system enforce those genre and tempo rules automatically. Finally, Pay to Request is tested with a \$0.99 boost option offered during the morning rush, and because the prototype has no bank account behind it, this last feature is only partially demonstrated through the Stripe integration.

### 3.4 How (Scenario)

The scenario begins when a customer named Alex sits down, notices the QR card on the table, and scans it with a phone. The Rhythmiq portal opens directly in the browser with no download needed, and Alex sees the current queue along with a search bar.

Alex searches for “Golden Hour” by JVKE. The Vibe Engine checks the song’s metadata against the café’s rules. Its genre is pop/indie, which is allowed, it is not flagged as explicit, and its tempo of 98 BPM is under the 120 BPM limit, so it passes every check and is added to the queue.

Alex then chooses to pay \$0.99 to boost the song higher in the queue, and the payment is processed automatically through Stripe. Throughout this process the staff receive zero interruptions, and the song plays through SoundMachine on the café’s speakers.

Later, during a slow period, Jamie logs into the Business Control Center on the tablet at the counter. Jamie reviews the queue, checks the live revenue from requests, of which the café keeps 67%, and adjusts the genre allowlist for the afternoon shift. If a disruptive request ever did slip through, the baristas could remove it from the queue themselves, although only Jamie can change the café’s settings.

This shows how the owner remains in charge of the venue’s atmosphere while the day-to-day work of handling requests is handled by the system.

*[Insert Figure 3 here]*

**Figure 3:** Case study scenario flow. Alex scans the QR code, searches for a song, the Vibe Engine validates it, the song is queued, SoundMachine plays it, and staff remain uninterrupted.

### 3.5 Future Expansion

If the Brewpoint case study succeeds, Rhythmiq has several natural directions for growth. The first is scaling to small cafe chains of five to ten locations, using a centralized Control Center that supports a separate vibe profile for each location. The second is expanding into adjacent markets where atmosphere matters and budgets are limited, including college dining halls, hotel lobbies, waiting rooms, fitness studios, and barbershops. A third possibility is a business-to-business SaaS licensing model aimed at venue management companies that oversee many properties. Finally, the team could add loyalty and gamification features, such as “vibe points” that customers earn when their requests stay in the queue, to encourage repeat engagement.

## 4. Glossary

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**API (Application Programming Interface):** A set of protocols and tools that allow different software applications to communicate with each other.

**ASCAP (American Society of Composers, Authors, and Publishers):** One of the main U.S. performing rights organizations, which licenses music for commercial use.

**BMI (Broadcast Music, Inc.):** Another major U.S. performing rights organization, which manages commercial music licensing on behalf of songwriters and publishers.

**Business Control Center:** Rhythmiq's dashboard interface that allows business owners and staff to manage song requests, set filters, track revenue, and control the atmosphere.

**Commercial License:** A legal permit that allows businesses to play music publicly in a commercial setting.

**Django:** A popular web framework for the Python programming language, used to build Rhythmiq's web application.

**Hardware Agnostic:** Software that can run on any existing device without requiring specific or proprietary hardware.

**Metadata:** Information, such as author, creation date, or file size, that describes a data point or data set [IBM, 2024]. For a song, this includes details such as genre, tempo, and mood.

**MFCD (Major Functional Components Diagram):** A diagram showing the main software components of a system and how they interact, as defined in CS 410.

**NRG (National Research Group):** A global insights and strategy firm specializing in the intersection of technology and entertainment.

**Pay-to-Request:** A feature allowing customers to pay to have their song moved higher in the queue.

**PostgreSQL:** An open-source database system used to store Rhythmiq's venue, song, and user data.

**QR Code (Quick Response Code):** A scannable two-dimensional barcode that provides quick access to a website or application when read by a smartphone camera.

**Revenue Stream:** A source of income for a business; in this case, the owner's 67% share of pay-to-request transactions.

**SESAC (Society of European Stage Authors and Composers):** Another U.S. performing rights organization that provides commercial music licensing.

**Soundcharts API:** An application programming interface that allows developers to integrate the Soundcharts music catalog and song metadata into their applications.

**SoundMachine:** A music streaming service that provides commercially licensed background music for businesses and handles audio playback for Rhythmiq.

**Stripe API:** A third-party payment service that Rhythmiq uses to process customer payments without storing card data itself.

**Vibe:** The mood of a place, situation, or person and the way it makes people feel [Cambridge Dictionary, 2021].

**Vibe Engine:** Rhythmiq's proprietary, in-house song selection and validation algorithm, which evaluates customer song requests against a venue's configured constraints before allowing them into the queue.

**Vibe Management:** The process of controlling and curating the musical atmosphere of a business.

**Web App:** A software application that runs in a web browser rather than being installed locally.

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